

Question 1**(23 marks)**

- (a) Simplify: 2
- i) $\sqrt{27}$
- ii) $\sqrt{72} - \sqrt{50}$
- (b) Expand and simplify: 2
- $(3\sqrt{2} + 1)(\sqrt{2} - 1)$
- (c) Evaluate $\sqrt{\frac{4.56 + 86.7}{6.4^3 \times 5.63}}$ correct to 3 significant figures. 2
- (d) Express $0.\dot{3}\dot{5}$ as a simple fraction. 3
- (e) Factorise completely: 2
- i) $4x^2 - 36$
- ii) $5y - 15 + xy - 3x$ 2
- (f) Simplify:
- i) $\frac{x-4}{2} + \frac{x+3}{3}$ 2
- ii) $\frac{a+2}{a^2-3a-4} \div \frac{a^2+4a+4}{a+1}$ 3
- (g) Solve the equation and leave the answer in simplest surd form: 3
- $n^2 + 10n + 7 = 0$
- (h) A number is increased by $7\frac{1}{2}\%$ to give 86. Find the number. 2

Question 2

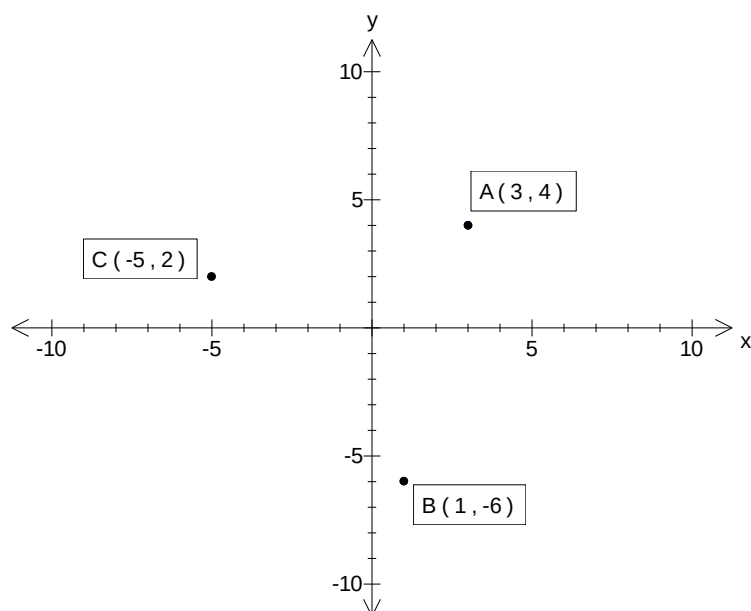
(21 marks)

- (a) Solve $|2x - 1| < 3$ and graph the answer on a number line. 3
- (b) Solve $2^x = 128$ 2
- (c) Solve simultaneously:
 $3x + 2y = -6$ 3
 $x - 2y = -10$
- (d) State the domain of the function:
 $y = \frac{4}{x - 3}$ 1
- (e) State the range of the function:
 $y = 3^x + 3$ 1
- (f) Determine $f(-x)$ for the function $f(x) = x^3 - 5x$ and hence state whether the function is **odd**, **even** or **neither**. Give a **reason** for your answer. 2
- (g) Simplify $8^{\frac{-2}{3}} \times 49^{\frac{1}{2}}$ 3
- (h) Simplify $|-5| - |8|$. 2
- (i) i) Sketch the graphs of $y = 2x + 3$ and $y = x^2$ on the same Cartesian Plane. Clearly show the intercepts with the axes. 3
- iii) Shade the region of the plane defined by the inequations $y \geq x^2$ and $y \leq 2x + 3$. 1

Question 3

(22 marks)

(a)



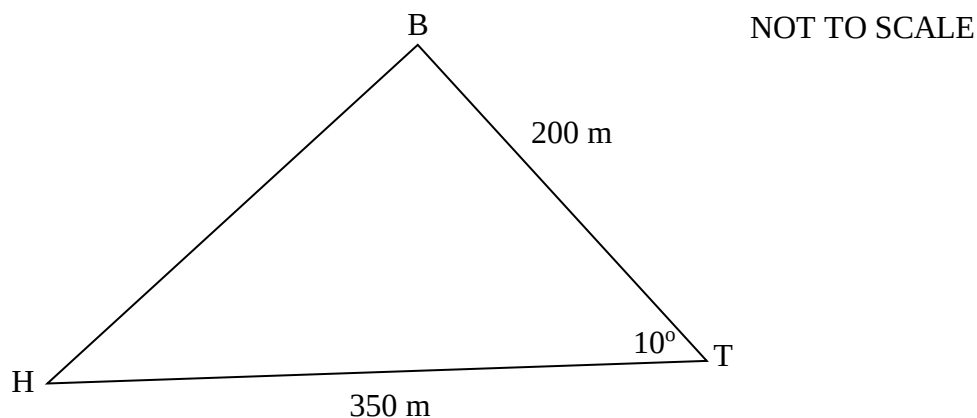
The points $A(3,4)$, $B(1,-6)$ and $C(-5, 2)$ are the vertices of a triangle.

- i) Show that the equation of the line AC is $x - 4y + 13 = 0$.
Call this equation line k . 3
 - ii) $P(2, -1)$ is the mid-point of AB . Find the gradient of BC and hence, find the equation of the line l through P parallel to BC . 3
 - iii) Find the point of intersection Q of the lines k and l . 3
 - iv) Show that Q is the mid-point of AC . 2
 - v) Show that $PQ = \frac{1}{2} BC$ 4
 - vi) Find the perpendicular distance from B to AC . 2
- (b) Find the **length of radius** and the **coordinates of the centre** of the circle: 2

$$(x - 4)^2 + (y - 5)^2 = 16$$
- (c) Find the equation of the line with x -intercept 4 that makes an angle of 45° with the x -axis. 3

Question 4**(15 marks)**

- (a) Solve $2\cos \theta = -1$ for $0^\circ \leq \theta \leq 360^\circ$ 3
- (b) Show that $\sec \theta + \tan \theta = \frac{1 + \sin \theta}{\cos \theta}$ 3
- (c) Find the exact values of:
- i) $\operatorname{cosec} 135^\circ$ 2
- ii) $\tan(-150^\circ)$ 2
- (d) If $\cos \alpha = \frac{2}{3}$, find the exact value of $\tan \alpha$ where α is acute. 2
- (e) On a golf course, the distance from a tee T to the hole H is 350 m. A golfer's ball comes to rest at point B , 200 m from T . Angle HTB is 10° , as shown in the diagram. How far is B from H ? Give your answer correct to 2 decimal places. 3



Question 5

(21 marks)

a) A series has n th term given by $T_n = n^3 - 5$. Find:

i) the 4th term

2

ii) which term is 5827

3

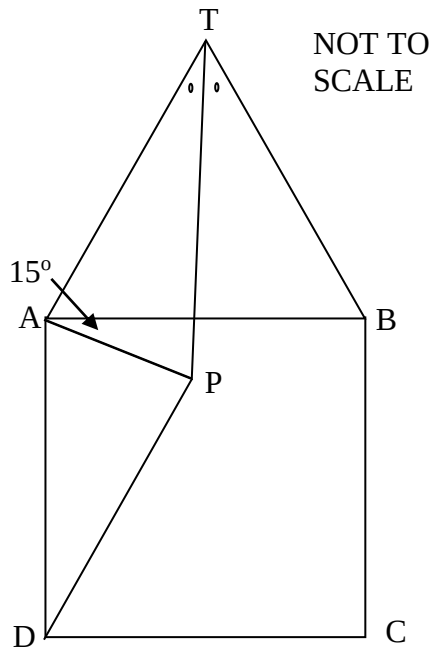
b) The limiting sum of a geometric series is $-\frac{3}{10}$ and its first term is $-\frac{1}{2}$. Find the common ratio of the series.

3

c) Find which term -370 is in the series $17 + 8 - 1 - \dots$

3

d) In the diagram, $ABCD$ is a square and ABT is an equilateral triangle. The line TP bisects $\angle ATB$, and $\angle PAB = 15^\circ$.



i) Copy the diagram onto your examination paper and explain why $\angle PAT = 75^\circ$

3

ii) Prove that $\triangle TAP \equiv \triangle DAP$

3

iii) Find $\angle APT$.

1

iv) Prove that $\triangle DAP$ is isosceles.

3

Q. a) (i) $\sqrt{27} = \sqrt{3 \times 3 \times 3} = 3\sqrt{3}$

(ii) $\sqrt{12} - \sqrt{50} = 2\sqrt{3} - 5\sqrt{2}$
 $= \sqrt{2}$ ✓

(b) $(3\sqrt{2} + 1)(\sqrt{2} - 1)$
 $= 3 \cdot 2 + \sqrt{2} - 3\sqrt{2} - 1$
 $= 6 - 2\sqrt{2} - 1$
 $= 5 - 2\sqrt{2}$ ✓

(k) $\sqrt{\frac{4.56 + 86.7}{6.4^3 \times 5.63}}$
 $= 0.249$ ✓ (3.s.f.)

d. Let $x = 0.3535$
 $100x = 35.3535$ ✓
 $- x = 0.3535$ ✓
 $99x = 35$
 $x = \frac{35}{99}$ ✓

e(i) $4x^2 - 36 = 4(x^2 - 9)$ ✓
 $= 4(x-3)(x+3)$ ✓ (2)

(ii) $5y - 15 + xy - 3x$
 $= 5(y-3) + x(y-3)$ ✓
 $= (y-3)(5+x)$ ✓

(f) (i) $\frac{x-4}{2} + \frac{x+3}{3}$
 $\frac{3(x-4)}{6} + \frac{2(x+3)}{6}$ ✓
 $= \frac{3x-12+2x+6}{6}$ ✓
 $= \frac{5x-6}{6}$ ✓

(2) f(ii) $\frac{a+2}{a^2-3a-4} \div \frac{a^2+4a+4}{a+1}$

$= \frac{(a+2)\cancel{\sqrt{a-4}}}{(a-4)(a+2)} \times \frac{(a+1)\cancel{\sqrt{a+2}}}{(a+2)(a+2)}$
 $= \frac{1}{(a-4)(a+2)}$ ✓ (3)

(2) (g) $n^2 + 10n + 7 = 0$
 cannot factorise
 $\therefore$ use formula

$a=1$
 $b=10$
 $c=7$

$n = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{-10 \pm \sqrt{10^2 - 4(1)(7)}}{2(1)}$ ✓

$= \frac{-10 \pm \sqrt{100 - 28}}{2}$ ✓

$= \frac{-10 \pm \sqrt{72}}{2}$ ✓ (3)

$\therefore n = \frac{-10 + \sqrt{72}}{2}$ or $n = \frac{-10 - \sqrt{72}}{2}$
 $\Rightarrow -5 \pm 3\sqrt{2}$

(2) (h) $x \times 1.075 = 86$
 $\therefore x = \frac{86}{1.075}$ ✓
 $= 80$ ✓ (2)

[23]

(2)

Q2

$$(a) |2x-1| < 3$$

$$2x-1 < 3 \quad \text{or} \quad -(2x-1) < 3$$

$$2x < 4 \quad \checkmark \quad -2x+1 < 3 \quad \checkmark$$

$$x < 2 \quad \checkmark \quad -2x < 2 \quad \checkmark$$

$$x > -1$$

$$\therefore -1 < x < 2 \quad \checkmark$$



$$(b) 2^x = 128$$

$$2^x = 2^7 \quad \checkmark$$

$$x = 7 \quad \checkmark$$

$$(c) \begin{cases} 3x+2y = -6 & \text{--- (A)} \\ x-2y = -10 & \text{--- (B)} \end{cases}$$

$$A+B \quad 4x = -16 \quad \checkmark$$

$$x = -4 \quad \checkmark$$

Substitute $x = -4$ into (A)

$$3(-4) + 2y = -6 \quad \checkmark$$

$$2y = -6 + 12 \quad \checkmark$$

$$2y = 6 \quad \checkmark$$

$$y = 3 \quad \checkmark$$

$$\therefore x = -4 \text{ and } y = 3$$

$$(d) y = \frac{4}{x-3} \quad x \neq 3 \quad \checkmark$$

$$\therefore D = \{x : x \neq 3, \text{ all real values}\} \quad (1)$$

$$(e) R = \{y : y \text{ all real values } y > 3\} \quad (2)$$

$$(f) f(x) = x^3 - 5x$$

$$\therefore f(-x) = (-x)^3 - 5(-x)$$

$$= -x^3 + 5x \quad \checkmark$$

$$= -(x^3 - 5x)$$

$$f(-x) = -f(x) \quad \checkmark$$

$$\therefore \text{odd} \quad \checkmark$$

(2)

$$(3) (g) 8^{-\frac{2}{3}} \times 49^{\frac{1}{2}}$$

$$= \frac{1}{8^{\frac{2}{3}}} \times (7^2)^{\frac{1}{2}} \quad \checkmark$$

$$= \frac{1}{\sqrt[3]{8^2}} \times 7 \quad \checkmark$$

$$= \frac{1}{\sqrt[3]{(2^3)^2}} \times 7 \quad \checkmark$$

(2)

$$= \frac{1}{\sqrt[3]{2^6}} \times 7 \quad \checkmark$$

$$= \frac{1}{2^2} \times 7 \quad \checkmark$$

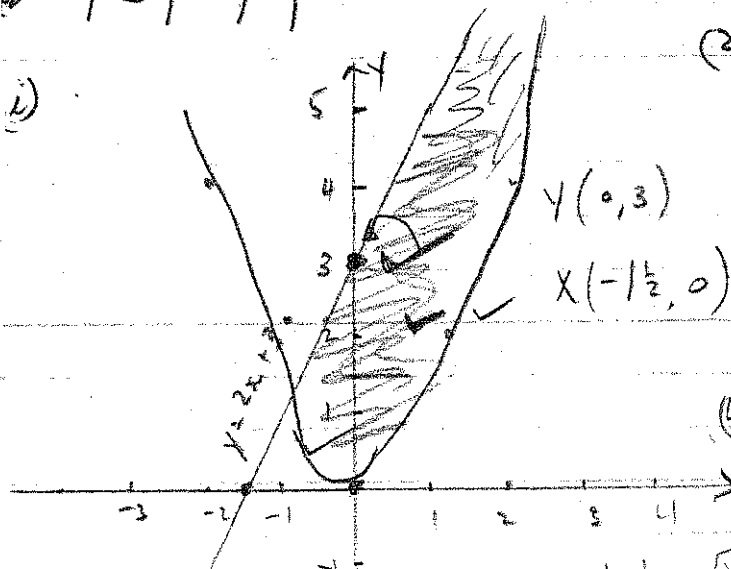
$$= \frac{1}{4} \times 7 \quad \checkmark$$

$$= 7/4$$

(3)

$$(b) |-5| - |8| = 5 - 8 = -3 \quad \checkmark \quad \checkmark \quad \checkmark$$

(1)



(2)

(4)

Total: [2]

$$3(i) \quad m = \frac{4-2}{3-(-5)} = \frac{2}{8} = \frac{1}{4} \checkmark$$

$$\therefore y - 4 = \frac{1}{4}(x - 3) \checkmark$$

$$4y - 16 = x - 3 \checkmark$$

$$0 = x - 4y + 13 \quad (3)$$

$$(ii) \quad P(2, -1), \parallel \text{ to } BC$$

$$\text{Gradient at } BC = \frac{-6-2}{1-(-5)}$$

$$= \frac{-8}{6} \checkmark$$

$$= -4/3$$

$$\therefore \text{Gradient at } l = -\frac{4}{3}$$

$$y - (-1) = -\frac{4}{3}(x - 2)$$

$$y + 1 = -\frac{4}{3}(x - 2) \checkmark$$

$$3y + 3 = -4x + 8 \quad (3)$$

$$4x + 3y - 5 = 0 \quad \checkmark \text{ or } y = -\frac{4}{3}x + \frac{5}{3}$$

$$(iii) \quad k: x - 4y + 13 = 0 \quad \text{--- (A)}$$

$$l: 4x + 3y - 5 = 0 \quad \text{--- (B)}$$

$$\textcircled{A} \times 4 \quad -4x + 16y + 52 = 0 \quad \text{--- (C)}$$

$$\textcircled{B} + \textcircled{C}$$

$$19y - 57 = 0$$

$$19y = 57 \checkmark$$

$$y = 3$$

$$x - 4(3) + 13 = 0 \checkmark$$

$$x - 12 + 13 = 0$$

$$x = -1 \checkmark$$

$$\therefore Q(-1, 3)$$

(3)

(iv)

$$AC \text{ Midpoint} = \left(\frac{3+(-5)}{2}, \frac{4+2}{2} \right)$$

$$= \left(-\frac{2}{2}, \frac{6}{2} \right)$$

$$= (-1, 3) \checkmark$$

$$\therefore Q(-1, 3) \text{ is } (2)$$

Midpoint at AC

$$(v) \quad PQ = \frac{1}{2} BC$$

$$P(2, -1)$$

$$Q(-1, 3)$$

$$B(1, -6)$$

$$C(-5, 2)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$PQ_d = \sqrt{(-1-2)^2 + (3-(-1))^2} \checkmark$$

$$= \sqrt{9 + 16} \checkmark$$

$$= \sqrt{25} \checkmark$$

$$= 5 \checkmark$$

$$BC_d = \sqrt{(-5-1)^2 + (2-(-6))^2}$$

$$= \sqrt{36 + 64} \checkmark$$

$$= \sqrt{100} \checkmark$$

$$= 10 \checkmark$$

$$\therefore PQ = \frac{1}{2} BC$$

(4)

$$(vi) \quad x - 4y + 13 = 0 \quad (AC)$$

$$a=1, b=-4, c=13 \quad B(1, -6)$$

$$d = \frac{|(1)(1) + (-4)(-6) + 13|}{\sqrt{1^2 + (-4)^2}} \checkmark$$

$$= \frac{|1 + 24 + 13|}{\sqrt{17}} = \frac{38}{\sqrt{17}} \text{ units} \checkmark$$

(2)

Q3(b) $(x-4)^2 + (y-5)^2 = 16$

$r = \sqrt{16}$

$r = 4$

Coordinates of centre $(4, 5)$ (2)

(c) $X(4, 0)$

$M = \tan 45^\circ$

$\therefore M = 1$

$\therefore y = 1x + b$

Substitute $X(4, 0)$:

$0 = 1(4) + b$

$-4 = b$

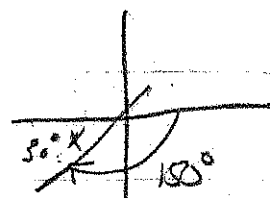
$\therefore y = 1x - 4$

[Total: 22] (2)

(ii) $\tan(-150^\circ)$

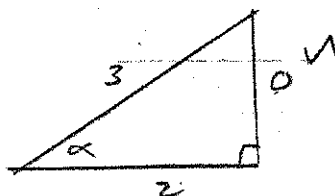
$= \tan 30^\circ$

$= \frac{1}{\sqrt{3}}$



(2)

(d) $\cos \alpha = \frac{2}{3} \left(\frac{a}{h} \right)$



$a^2 = 3^2 - 2^2$

$= 9 - 4$

$= 5$

$a = \sqrt{5}$

$\tan \alpha = \frac{\sqrt{5}}{2}$

(2)

(e) $t^2 = b^2 + h^2 - 2bh \cos T$
 $= (350)^2 + (200)^2 - 2(350)(200) \cos 10^\circ$

$= 24626.9$

$t = 156.93 \text{ m}$

(3)

[Total: 15]

Q4

(a) $2 \cos \theta = -1$

$\cos \theta = -\frac{1}{2}$

θ is in 2nd + 3rd quadrant

2nd quadrant
 $180^\circ - 60^\circ$

$= 120^\circ$

3rd quadrant
 $180^\circ + 60^\circ$

$= 240^\circ$

(3)

(b) $\sec \theta + \tan \theta = \frac{1 + \sin \theta}{\cos \theta}$

LHS = $\sec \theta + \tan \theta$

$= \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}$

$= \frac{1 + \sin \theta}{\cos \theta}$

(3)

$= \text{RHS}$

(c) (i) $\csc 135^\circ = \frac{1}{\sin 135^\circ}$

$= \frac{1}{\sin(180^\circ - 45^\circ)}$

$= \frac{1}{\sin 45^\circ}$

$= \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$

(2)

Q5

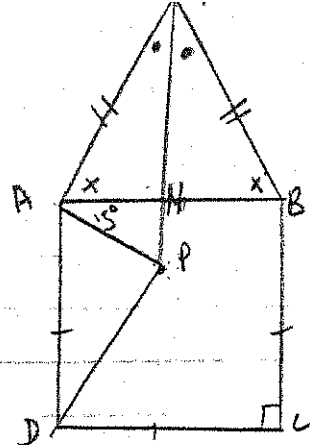
9.2) $T_n = n^3 - 5$

$T_4 = 4^3 - 5$ ✓

$= 59$ ✓

92) $T_n = n^3 - 5$
 $T_4 = 4^3 - 5$ ✓
 $= 59$ ✓

(a)



(iii) $T_n = 5827 \quad n = ?$
 $5827 \checkmark = n^3 - 5 \checkmark$
 $5832 \checkmark = n^3 \checkmark$
 $\sqrt{5832} = n \checkmark$
 $18 = n \checkmark$

$$S_{\infty} = \frac{a}{1-r}$$

$$a = -\frac{1}{2}$$

$$S_{\infty} = -\frac{3}{10}$$

$$\frac{-\frac{3}{10}}{1-r} = -\frac{1}{2}$$

$$\begin{aligned} -3(1-r) &= -5 \checkmark \\ -3 + 3r &= -5 \checkmark \\ 3r &= -2 \\ r &= -2/3 \checkmark \end{aligned}$$

(c) $17 + 8 + -1 - \dots$ $T_n = -370$

$$\begin{aligned} a &= 17 & d &= -9 \\ T_n &= a + (n-1)d \\ -370 &= 17 + (n-1)(-9) \checkmark \\ \checkmark -370 &= 17 - 9n + 9 \\ -370 - 26 &= -9n \quad \checkmark \\ \checkmark -396 &= -9n \\ \checkmark \frac{396}{9} &= n \quad \checkmark \\ 44 &= n \quad \checkmark \end{aligned}$$

(3)

$$\begin{aligned} \text{(i)} \quad \angle TAB &= 60^\circ \checkmark \text{ (TAB equilateral)} \\ \angle PAT &= \angle TAB + \angle PAB \\ &= 60^\circ + 15^\circ \checkmark \\ &= 75^\circ \checkmark \end{aligned} \quad (3)$$

(ii) $\angle DAP = 90^\circ - 15^\circ$ ($\angle DAB$ is 90°)
 $= 75^\circ \checkmark$

$TA = AB$ ✓ (sides of equilateral)
 but $AB = AD$ (sides of square)
 ∴ $TA = AD$ ✓

In ΔTAP and ΔDAP
AP is common ✓

$$\angle PAT = \angle DAP \text{ (both } 75^\circ)$$

$$TA = DA \quad \checkmark$$

$$\therefore \Delta TAP \equiv \Delta DAP \quad (SAS) \quad (3)$$

(iii)

$$\angle ATP = 30^\circ \quad (\text{Half of } \angle ATB)$$
$$\angle APT = 180^\circ - (75^\circ + 30^\circ)$$
$$= 75^\circ \quad (2)$$

(iv) $\triangle ATP$ is isosceles ✓
 $(\angle PAT = \angle TAP = 75^\circ)$
 Since $\triangle ATP \cong \triangle DAP$ ✓
 $\triangle DAP$ is also ✓ (3)
 isosceles [22]

22

